

0 3 . 1

Describe the relationship between size and surface area to volume ratio of organisms.
[1 mark]

0 3 . 2

A scientist calculated the surface area of a large number of frog eggs. He found that the mean surface area was 9.73 mm^2 . Frog eggs are spherical.

The surface area of a sphere is calculated using this equation

$$\text{Surface area} = 4\pi r^2$$

where r is the radius of a sphere

$$\pi = 3.14$$

Use this equation to calculate the mean diameter of a frog egg.

Show your working.

[2 marks]

Diameter = _____ mm



The scientist calculated the ratio of surface area to mass for eggs, tadpoles and frogs. He also determined the mean rate of oxygen uptake by tadpoles and frogs.

His results are shown in **Table 2**.

Table 2

Stage of frog development	Ratio of surface area to mass	Mean rate of oxygen uptake / $\mu\text{mol g}^{-1} \text{h}^{-1}$
Egg	2904 : 1	no information
Tadpole	336 : 1	5.7
Adult	166 : 1	1.3

0 3 . 3 The scientist used units of $\mu\text{mol g}^{-1} \text{h}^{-1}$ for the rate of oxygen uptake.

Suggest why he used μmol in these units.

[1 mark]

0 3 . 4 The scientist decided to use the ratio of surface area to mass, rather than the ratio of surface area to volume. He made this decision for practical reasons.

Suggest **one** practical advantage of measuring the masses of frog eggs, tadpoles and adults, compared with measuring their volumes.

[1 mark]

Turn over ►



0 3 . 5

Explain why oxygen uptake is a measure of metabolic rate in organisms.

[1 mark]

0 3 . 6

A student who looked at these results said that they could not make a conclusion about the relationship between stage of development and metabolic rate.

Use information in **Table 2** to explain reasons why they were unable to make a conclusion.

[3 marks]

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